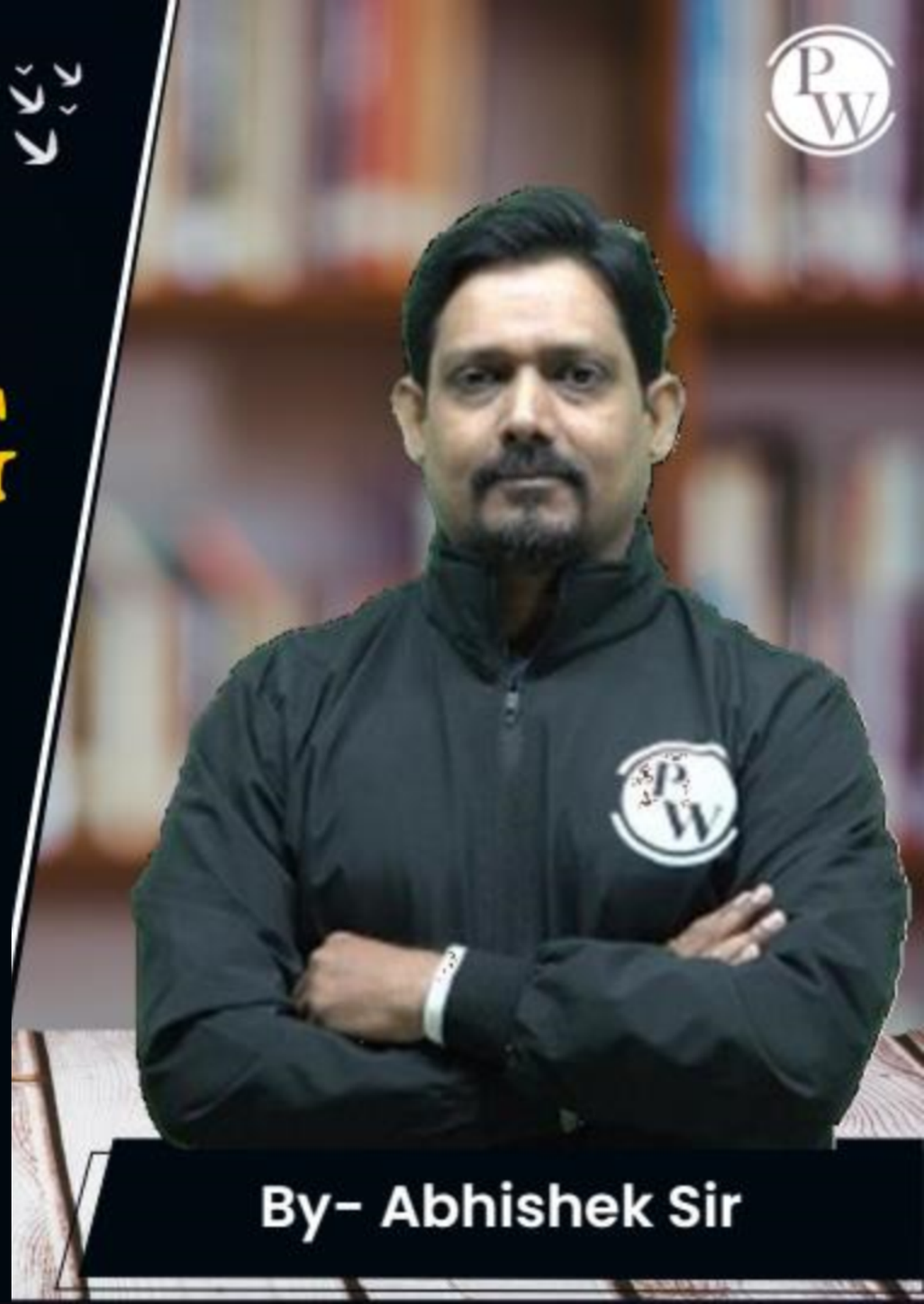
CS & IT ENGINEERING

Data Structure & Programming

Array

DPP 01 Discussion Notes



[NAT]

#Q. Consider a 1D array a with 9 elements. If the base address of the array is 108 and the size of each array element is 4 bytes, the address of a[7] is- _____
(Assume array index starts from 0)

BA + i x Size

108 + 7 x 4 = 108 + 28 = 136

[NAT]

#Q. Consider a 1D array a[-127.....,+255] where -127 and +255 are the starting index and ending index of the array respectively. The number of elements in the array is _____.

LB — UB

→ UB - LB + 1

255 - (-127) + 1

= 255 + 127 + 1 = 383

[NAT]

#Q. Consider a 1D array a with 19 elements. If the base address of the array is 108 and the size of each array element is 4 bytes, the address of a[7] is- (Assume array index starts from -11)_____

$$A[7] = BA + (i - LB) \times Size$$

$$108 + (7 - (-11)) \times 4B$$

$$108 + 18 \times 4 = 108 + 72$$

$$180$$

[NAT]

#Q. Consider a 2D array a[-127 to +255][-13 to +14]. The number of elements in the array is 10724.

LB_1 UB_1 LB_2 UB_2

No. of element Rows : 255 − (−127) + 1

383 Rows

No. of element in column 14 − (−13) + 1 = 28 Columns

383 × 28

3064

766X

10724

[NAT]

#Q. Consider the natural numbers from 1 to 256 are stored in a 2D array arr[-28 to 3][-3 to 3]. Find the element present at location arr[-16][1]. (Suppose, the elements are stored in row-major order) 89

1 Base Addres A[-16][1]

$$1 + [(-16 - (-28)) \times (3 - (-3) + 1) + (1 - (-3))] \times 1$$

$$1 + [12 \times 7 + 4] \times 1 = 89$$

[NAT]

#Q. Consider the natural numbers from 1 to 256 are stored in a 2D array arr[-28 to 3][-3 to 3]. Find the address of the location arr[-16][1] if the starting address of the array is 625 and size of each element is 4 bytes. (Suppose, the elements are stored in row-major order)

$$625 + [(-16 - (-28)) \times (3 - (-3) + 1) + (1 - (-3))] \times 4$$

$$625 + 88 \times 4 = 625 + 352 = 977$$

[NAT]

#Q. Consider the whole numbers from 0 to 127 are stored in a 2D array arr[0 to 15][0 to 7]. Find the element present at location arr[6][4]. (Suppose, the elements are stored in column-major order) 70

$$0 + [(4-0) \times 16 + (6-0)] \times 1$$

$$0 + [64 + 6] = 70$$

[NAT]

#Q. Consider a 2D array arr[-15 to 15][-7 to 7]. Find the address of the location arr[-1][5] if the starting address of the array is 500 and size of each element is 4 bytes. (Suppose, the elements are stored in column-major order) 2044.

$$Arr[-1][5]$$

$$500 + [(5-(-7)) \times (15-(-15)+1) + (-1-(-15)] \times 4$$

$$500 + [12 \times 31 + 14] \times 4$$

$$500 + [372 + 14] \times 4 = 2044$$

THANK - YOU